ALFRED OJUKU OKINYI

BACKEND DEVELOPER | DATA ENGINEER

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PROFESSIONAL SUMMARY

Backend-focused Data Engineer with 3+ years of experience designing relational data models, building production REST APIs, integrating enterprise and payment systems, and operating cloud-hosted services. Hands-on with Python, FastAPI, SQLAlchemy, PostgreSQL, MySQL, Redis, Docker, Linux and Google Cloud. Experienced in taking business requirements from schema design through API implementation, deployment, troubleshooting and operational support. Strong record of translating complex workflows into maintainable systems while reducing cost and improving reliability and data quality.

CORE SKILLS

Backend: Python, FastAPI, REST API design, SQLAlchemy, Pydantic, asynchronous services, webhooks
Databases: PostgreSQL, MySQL, BigQuery, relational modelling, schema design, migrations, query optimization
Cloud and Infrastructure: Google Cloud Run, Cloud SQL, Pub/Sub, GCS, Docker, Linux, Nginx, Redis
Security and Delivery: RBAC, JWT/OAuth concepts, input validation, rate limiting, audit logging, Git/GitLab, CI/CD
Data and Integration: Business Central 365 APIs, M-Pesa, Freshdesk, Gmail API, dbt, Apache Spark, Airbyte
Engineering Workflow: API testing, production debugging, logging/monitoring, AI-assisted coding and documentation

PROFESSIONAL EXPERIENCE

Data Engineer | *Roam Electric - Nairobi, Kenya* **Apr 2025 - Present**

- Design and maintain backend services in FastAPI for RoamCore, supporting asset, customer, service, payment and telemetry workflows across internal applications.
- Model relational schemas and implement business logic across PostgreSQL and MySQL, including multi-entity relationships, JSON data, migrations, triggers and cross-system synchronization.
- Build secure REST endpoints and integrations for Microsoft Dynamics 365 Business Central, M-Pesa, Freshdesk and Gmail/Pub/Sub, including webhooks, callbacks and error-handling workflows.
- Deploy and operate containerized services on Google Cloud Run and Linux infrastructure using Docker, Cloud SQL, Pub/Sub, GCS, Redis and Nginx; diagnose failures using application and platform logs.
- Develop role-based access workflows and contribute to API protection through authentication, authorization, validation, rate-limiting and auditability controls.
- Support Git-based release workflows and CI/CD across development, staging and production; perform API load testing and tune database connection pools for production workloads.

Junior Data Engineer | *Roam Electric - Nairobi, Kenya* **Oct 2023 - Apr 2025**

- Built the company's asset, customer and telemetry data foundation, establishing trusted operational datasets and integration points for business applications.
- Developed and deployed M-Pesa payment endpoints that improved payment processing and reconciliation for Roam Hubs.
- Created custom Business Central API pages, actions and reports to support operational processes and reduce reliance on external consultants.
- Built ETL pipelines from Business Central and Freshdesk into the data warehouse using Python, SQL, dbt, Airbyte and Spark, with dimensional models for analytics.
- Reduced monthly Google Cloud spend from approximately USD 1,500 to USD 700 through infrastructure and workload optimization.
- Customized Frappe CRM and developed web pages that enabled sales agents to create and track leads more effectively.

Process Engineer Apprentice | *Roam Electric - Nairobi, Kenya* **Apr 2023 - Oct 2023**

- Analyzed production and operational processes, built reporting tools and automated recurring workflows, creating the foundation for a move into data and backend engineering.

SELECTED TECHNICAL PROJECTS

RoamCore Backend Services: Implemented FastAPI services and database models for ticketing, service-order, payment and user-management workflows, integrating internal systems with Business Central and external APIs.

Telemetry Platform: Supported ingestion and processing of high-volume motorcycle telemetry through GCP services, BigQuery, GCS and Spark, including operational troubleshooting and data-quality improvements.

Cloud and Release Engineering: Containerized and deployed Python services, configured reverse proxies and CI/CD environments, and supported structured feature-to-release workflows across dev, staging and production.

EDUCATION

Bachelor of Engineering in Electrical and Electronics Engineering | *Dedan Kimathi University of Technology - Nyeri, Kenya* **2018 - 2023**
Second Class Honours, Upper Division